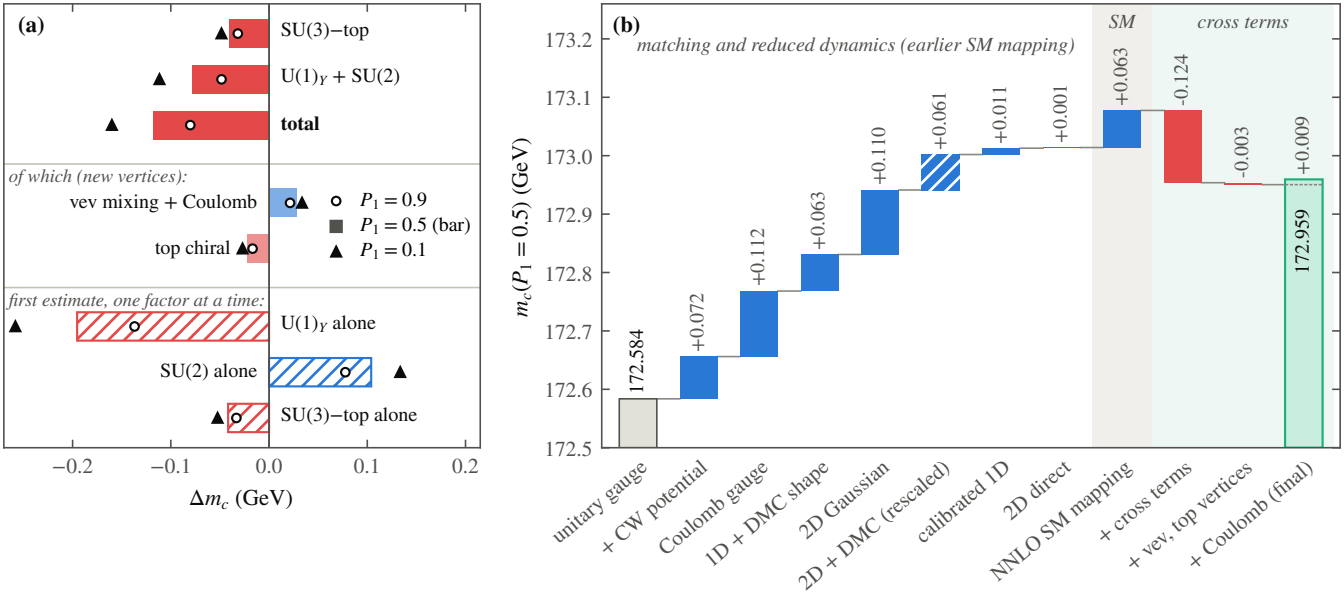
(a)
SU(3)−top
U(1)$_Y$ + SU(2)
total
of which (new vertices):
vev mixing + Coulomb
top chiral
$P_1 = 0.9$
$P_1 = 0.5$ (bar)
$P_1 = 0.1$
first estimate, one factor at a time:
U(1)$_Y$ alone
SU(2) alone
SU(3)−top alone
−0.2
−0.1
0.0
0.1
0.2
Δm_c (GeV)
(b)
matching and reduced dynamics (earlier SM mapping)
SM
cross terms
$m_c(P_1 = 0.5)$ (GeV)
173.2
173.1
173.0
172.9
172.8
172.7
172.6
172.5
172.584
+0.072
+0.112
+0.063
+0.110
+0.061
+0.011
+0.001
+0.063
-0.124
-0.003
+0.009
172.959
unitary gauge
+ CW potential
Coulomb gauge
1D + DMC shape
2D Gaussian
2D + DMC (rescaled)
calibrated 1D
2D direct
NNLO SM mapping
+ cross terms
+ vev, top vertices
+ Coulomb (final)